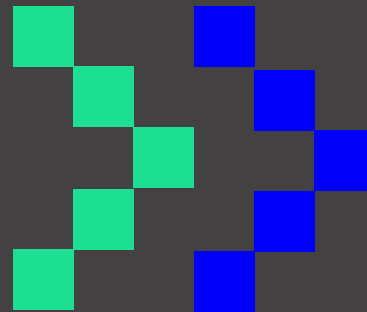


ECHO MICROSCOPES REVOLUTION

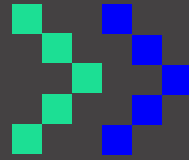


User Manual

ECHO Support
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ECHO's Mission Statement

ECHO is built on two foundations:

We build the best microscopes in the industry. With progressive design, we empower our users to change the way they view science.

We commercialize our products by providing unmatched value to our customers and distributors. Our brand changes outdated perceptions and defines the future of microscopy.

For more information about your Revolution, or any help with Setup and Operation, please contact us on our website at discover-echo.com.

2. Operating Instructions

2.1 Restrictions

Both the Revolution and the Revolution Software(the “Software”, “Revolution”, “App”) are intended for research, investigational, and educational usage only. The Revolution Microscope and Revolution Software are not intended to be used for diagnostic purposes, testing, or treatment in humans.

2.2 Maintenance

2.2.1 Hardware Maintenance

The Revolution requires minimal physical maintenance. In the case of an error or problem with your system, the included Computer and Revolution Microscope are both covered by a One-Year Manufacturer Warranty, starting 14 days after shipment from ECHO. Additional Warranty coverage, including an annual PM for GMP requirements, can be purchased from ECHO.

Your Revolution can be cleaned with up to a 70% Ethanol solution should it need to be decontaminated for experiments. Avoid spraying the solution directly onto the system to reduce the risk of seepage into the internal electronics.

There are only 4 cables that need to be plugged in to your Revolution and Computer – each device has a power cable, then there is a USB connection between your microscope and computer. Finally, if you purchased a

joystick, this would plug into your microscope. These cables are covered by the same One-Year Manufacturer Warranty.

2.2.2 Software Maintenance

As of publication, the ECHO Revolution comes with a Windows 10-based PC. Security (Build) updates to Windows 10 have not been known to cause any errors with the Revolution Software. However, it is recommended to turn off automatic Windows updates in the rare event that a Build update does interfere with a core function in the Revolution Software.

If you would like to update your Revolution Software, please contact your Local ECHO Rep and ECHO Support for help and advice on which software version may be right for you!



Img 1: Picture of the various ports on the Microsoft Surface Studio. Note that the right-most (SuperSpeed) USB port, Ethernet, and Power are highlighted.

Discover Echo disclaims all warranties and shall in no event be liable for any kind of damages caused by or arising out of any kind of operation or use in violation with the above safety and handling instructions.



2.3 Safety Instructions



This symbol means that there is risk of an electric shock. An electric shock could cause death or personal injury.

The Revolution Incubator has been certified by CA, EU, UK, and US safety standards and is safe to use when operated in accordance with this manual. Proper usage of the Revolution will limit the chance of an electric shock to the user.

Do not look directly into the LED illuminated light source. High strength LED light, particularly that from the UV and IR portions of the spectrum, can be damaging to your eyes. When opening user access panel to swap LEDs and Filter Cubes, ensure that the LED indicator is off in the Revolution Software, or close the App to turn off the microscope.

To reduce risk of injury, please use this system with care and take the following precautions:

- Only use the system once you have received training from an ECHO Representative, ECHO-authorized Representative and/or read this manual.
- Do not utilize any power bricks or cables that have not been supplied by Discover Echo (ECHO).

- Plug the system into an outlet that can be easily accessed to unplug the system from the wall.
- Do not open the system while the LEDs are still on.

2.3.1 Revolution Incubator

If you have purchased the Revolution Incubator, there are additional safety precautions relating to the Revolution Incubator and the FC-5N Automatic CO2/Air Mixer. Both the Revolution Incubator and FC-5N Automatic CO2/Air Mixer will ship with a user guide and/or informational pamphlet with information on how to correctly use the equipment.

The Revolution Incubator has been certified by CA, EU, UK, and US safety standards and is safe to use when operated in accordance with this manual.

To reduce risk of injury, please use this system with care and take the following precautions:

- Only use the system once you have received training from an ECHO Representative, ECHO-authorized Representative, and/or read this manual and the associated FC-5N manual and Revolution Incubator Information Pamphlet.
- Do not use this equipment outside. The Revolution Incubator is designed for indoor use only.

- Do not use unauthorized attachments or accessories. Only attachments and accessories approved by ECHO have been tested to guarantee safe operation of the equipment.
- Do not attempt to modify or utilize this product in any manner not specified by the manufacturer.
- Do not use or store the incubator on a flammable surface.
- Carry the incubator with two hands or via a cart. The incubator can crack or become damaged if dropped.



A typical
REVOLUTION will
include:

1. One Revolution Body,
containing
 - A. One 5MP Color Camera.
 - B. One 5MP Mono Camera
 - C. USB A Port.
 - D. Power Port.
 - E. Joystick Port.
2. One Condenser (if purchased):
 - A. LWD Universal Condenser.
 - B. High-Resolution Universal
Phase Turret Condenser
 - C. Swing-out Condenser.
3. One XY Motorized Stage.
4. Plate-inserts (if purchased)
5. Objective Lenses (if purchased)
6. LED Light Cubes and Filter
Cubes (if purchased)
7. One Surface Studio
Computer with wireless
keyboard and mouse.
8. One Joystick (if purchased)
9. Revolution power cable,
brick, and local adapter.
10. Computer power cable, brick,
and local adapter.
11. USB A to USB A cable.

2.5 Installation

Your Revolution Microscope will be installed by your local ECHO Representative or a Representative of your ECHO-Approved Distributor. If this service is unavailable at your location, please contact Discover ECHO via chat at discover-echo.com or email at info@discover-echo.com. Include your location, serial number, and distributor.

Users can, however, install or change out accessories on their own without the need for a visit from an ECHO Representative.

2.5.1 Switching Objectives

The Revolution supports up to 6 objectives on the same nosepiece turret (INP). It is easier to swap objectives in the inverted configuration.

1. Rotate to the desired slot.
2. If there is already an objective in this location, remove the objective by turning counterclockwise.
3. Install your new objective by aligning the thread in the turret with that of your objective. Rotate clockwise until the objective base is seated firmly against the nosepiece.

Your objective will need to be updated within the Revolution software. Updating your selected objective is discussed in Section 4.2.1 – Per-Lens Settings.

2.5.2 Switching Condensers

To install either the High-Resolution or Swing-Out condenser, you will need to first remove the ELWD Universal Condenser. The ELWD Condenser comes pre-installed if you ordered it.

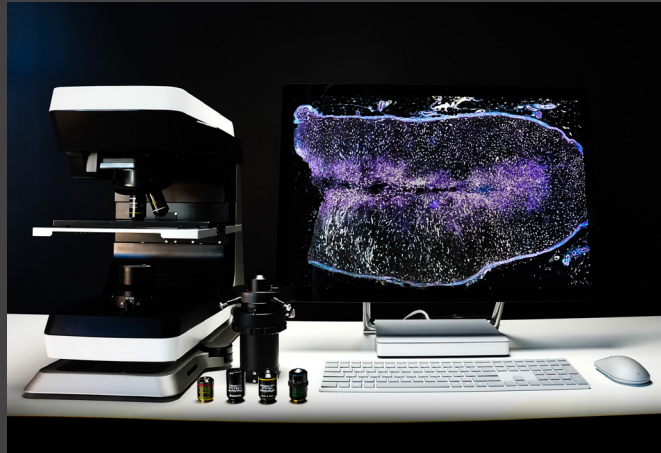
To install the High-Resolution Condenser (Upright Configuration Only):

1. Lower the condenser to the lowest possible position.
2. Install the condenser by sliding the base of the condenser with the phase selector facing the user.
3. Lock the condenser in place by sliding the locking mechanism into place.
4. Raise the condenser into position and begin Kohler Illumination Alignment.

To install the Swing-Out Condenser:

1. Lower the condenser to the lowest possible position.
2. Make sure the swing-out collimator is rotated to the down position before installing the condenser.
3. Lock the condenser in place by sliding the locking mechanism into place.
4. Flip the swing-out collimator into position.
5. Raise the condenser into position and begin Kohler Illumination Alignment.





Img 2: Revolution Microscope, Microsoft Surface Studio, with various available accessories. Not all accessories pictured will come with your scope. LED Light Cubes, Filter Cubes, Plate Inserts, and Objectives Lenses are all sold separately.

2.5.3 Swapping LED Cubes and Filter Sets

The Revolution supports up to 4 fluorescent channels to be installed at any time. To install a new channel:

1. Convert the Revolution into the inverted position.
2. Raise the nosepiece for clearance.
3. Remove the user access panel.
4. If there is not a position open, remove an LED and filter cube. The LED positions are marked on both the lightbox and cube turret.
5. Slot the LED cube into the desired channel. You will feel a magnetic attracting force guide the LED cube into position. Firmly press it to seat.
6. Slot the Filter cube into the corresponding channel number on the cube turret. You will feel a magnetic attracting force guide the Filter cube into position. Firmly press to seat.

2.5.4 Installing the Incubator

To install your ECHO Incubator onto your Revolution:

1. Place the microscope in the inverted position and center the stage with the supplied joystick.
2. Using two hands, slide the incubator onto the microscope.
3. Attach both magnetic covers (Rear and Condenser) to seal the incubator.

Plug in power to the incubator. Additional operating information, including how to attach the CO2 mode accessories, can be found in the informational pamphlet that ships with your incubator.



Img 3: Rear panel of the Revolution Microscope. The sticker with Serial Number and Wi-Fi Network (SSID and Password), along with the various ports are highlighted. From left-to-right, the ports are Joystick, Power, and USB.

In normal operation, the LEDs located between the Power and USB ports will be blinking green, off, and solid green.



3. System Initialization

3.1 First-Time Use

Your Revolution will be installed by your local ECHO Representative or a Representative of your ECHO-Approved Distributor. However, there are still a few steps you may need to complete:

1. Plug in the power cable for both your Revolution and Computer.
2. Plug in the USB A-to-A cable into both your Revolution and Computer.
 - a. If your Revolution shipped with the Microsoft Surface Studio, plug the cable into the right-most USB port as this is a USB 3.1 port and will provide the fastest data connection speed
3. If you have purchased a joystick, plug your joystick into your Revolution.
4. Power on your computer and log in.
5. Open the Revolution software on your computer. This will power-on and initialize the microscope.

The software may take up to a minute to load and initialize your Revolution. Once this process is complete, you will be ready to start imaging.

3.2 Subsequent Use

For future use, you will not need to plug in any power or data connection cables. Therefore, the process to ready your microscope for subsequent use is quite easy:

1. Turn on your Computer.
2. Log-in to your account.
3. Open the Revolution software.

Your microscope will take about a minute to boot-up. Once this process is complete, you will be ready to start imaging.

If you would like to open up a previously-run experiment, go to the menu in the top-left corner. Select “Open” then navigate to where your experiment was saved. Open the .sws file. Please note that if the associated images were moved or deleted, the experiment will not correctly open within the Revolution software.



4. Revolution Software

The Revolution software is designed to be intuitive yet powerful, sharing a similar look and feel to the ECHO Pro App used to control the Rebel and Revolve microscope platforms. However, as the Revolution is a fully automated and more customizable platform, there are some features which differentiate between Revolution and the ECHO Pro App.

Some software features require optional packages to utilize. These packages are:

1. Revolution HyperScan Software – includes the HyperScan menu and Focus Mapping.
2. Revolution Multi-Well Points Module – includes Tiling at Point Feature and Focus Mapping.

4.1 Basic Operating Commands

Focusing can be done by scrolling in and out with the mouse wheel. Clicking on the mouse wheel will toggle between fine, medium, and course focus. Holding the ctrl key while scrolling toggles super-coarse mode. The joystick only focuses in fine or super-fine increments.

To autofocus, right-click the Live View window. Autofocus is not a magic focus that will find the best focal plane, but rather, help you focus once you've gotten close to the highest-contrast focal plane.

4.2 Mode Selection Ribbon

At the top of your screen, there is a small ribbon with the options “Basic Mode”, “Multimode” and “Hyperscan” (Hyperscan only available if purchased). This is your Mode Selection bar and allows you to toggle between more basic capturing and customizable experiments.

4.3 Settings and Preferences

4.3.1 Per-Lens Settings

In both Basic Mode and Multimode, users have several settings that can be specified per-lens (Per-Lens Settings, Lens Automation Settings). These settings can be accessed by clicking on the objective icon.

4.3.1.1 Select and Calibrate Lenses

To change the selected objective, click on the ellipses. This will bring up a hamburger menu with a list of all the objectives supported by your Revolution. Objectives are grouped by magnification (1.25x, 2x, 4x, etc.). Click on the desired objective to assign it to that position on your intelligent nosepiece (INP). Although the objective is assigned, it will still need to be calibrated for both BF and FL imaging.

To calibrate your objective, click Advanced > Calibration tab. Ensure that you are in-focus and located on an area of your sample with a high-level of detail, then center this on your

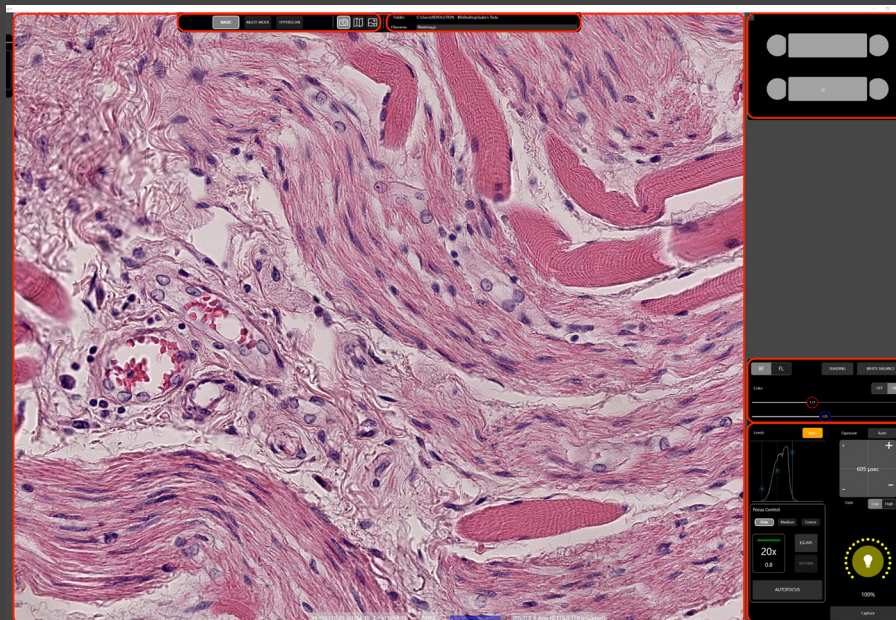
crosshairs. Click the Auto-Calibrate button. Revolution will automatically sample various points to determine the spatial calibration and orthogonality of the currently selected lens. If autocalibration fails, select another region of your sample and try again. If autocalibration repeatedly fails, contact ECHO Customer Support for assistance.

4.3.1.2 Shading Correction and White Balance (BF)

To white balance, click the White Balance button located next to the BF/FL toggle. This will automatically set the white balance, while the sliders will allow you to manually correct your white balance.

Img 4: Screen capture of the Revolution software, version 1.0.26, in BF Live View.

Various settings menus are highlighted, including the mode selector, stage insert manager, and lighting controls.



To set a shading correction, click the Shading Correction button located next to the White Balance button.

Both these features are also available underneath the Advanced > Calibration in the lens settings menu

When changing objectives and microscopy modes, it is important to ensure your Revolution's camera has undergone a shading correction. This will ensure an even illumination of the image. It is important to note that to perform an accurate white balance and shading correction, a region with no sample or sample holder, needs to be selected, as any sample, plastic, or glass could result in an improper white balance or shading correction. Ensure that your focal plane when setting a shading correction or white balancing is the same as your sample.

4.3.1.3 Shading Correction (FL)

The FL camera on the Revolution is a monochrome camera, thus eliminating the need for a White Balance. The only required software adjustment is a shading correction, which is done per-channel. To perform a shading correction in FL:

1. Ensure that the FL camera is selected.
2. Load a Chroma Autofluorescent slide. There are four slides corresponding to the 4 main color groups for fluorescent channels (Blue, Green, Yellow, Red). Ensure

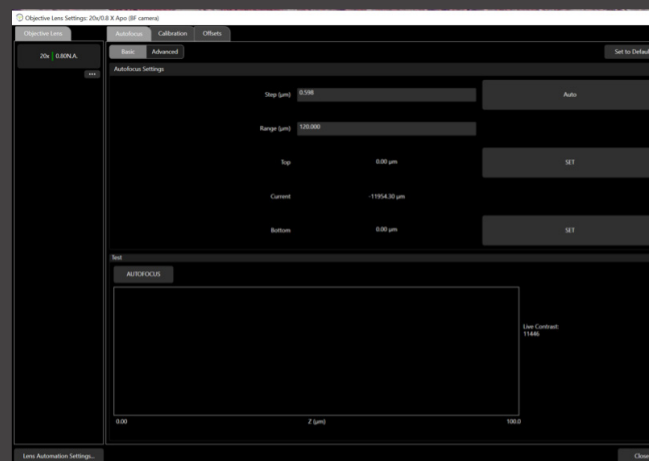
the proper slide is selected for your filter cube.

3. Click on the objective lens to open the lens settings menu.
4. Select Advanced > Set Shading Correction.
5. To use the shading correction, click Enable Shading Correction. If the dot is filled, the shading correction is enabled.

Note that TRANS does not require an Autofluorescent slide. Skip step 2 from the above directions when setting a shading correction for the TRANS channel.

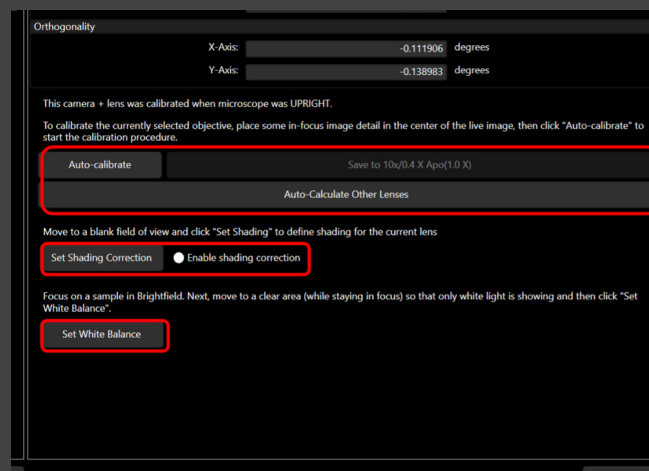


Img 6b: Screen capture of the image controls menu. In BF mode, the shading correction and white balance buttons are available in the top. In FL mode, these are not shown here.



Img 5: Screen capture of the Revolution software, version 1.0.26, in the Objective Lens Settings.

This menu can be reached by clicking on your objective lens, and allows for adjustment of the autofocus parameters, calibration of your lenses, and parfocality/parcentricity offsets between lenses..



Img 6a: Screen capture of the Objective Lens Calibration settings in Revolution Software version 1.0.26. The Calibration screen is open, and the buttons to auto-calibrate the orthogonality, set shading correction, and set the White Balance are all open.



4.3.2 Global Settings

To access the global settings, select the hamburger menu icon in the top corner of the software.

4.3.2.1 Open Results

To open previously completed experiments, click Open Results. This will allow you to open the .sws files created by the Revolution software.

4.3.2.2 Stage Insert Manager

This tab allows users to add additional stage inserts into the Revolution program. To add an additional stage, click “Add...” and upload a .png of your stage holder. You can change the length and width of your stage in mm (millimeters).

4.3.2.3 Microscope Initialization

Your Revolution microscope will initialize on startup unless disabled. Initialization will run through a short sequence of commands to re-home your stage, nosepiece, and light filter turret. Clicking this button will re-home your light filter turret.

4.3.2.4 Set User Preferences

There are several preferences that users can set to change the feel and responsiveness of their microscope. This manual will not go into depth on all settings, however, the most commonly adjusted settings will be explained in depth.

Underneath the general tab, there is an option to create a delay before

taking an image. The default is set to “10”, however, a larger delay may be recommended for if users are imaging live cells that are not adherent to your flask or plate. If there are white streaks when taking a tiled image, increasing this value is also recommended.

The focus tab allows you to change your multipliers for fine, medium, and coarse focus which are based on the Nyquist value for that objective. Note that this will only change the multipliers for your mouse wheel, and will not affect the super-fine focus on your Joystick. This will also not affect the multiplier for super-coarse focus (ctrl + scroll).

If you are running the same experiment repeatedly, it is recommended to turn on the auto-indexing feature. This will ensure that new experiments will not override a previous experiment by adding an auto-indexing tag to the end (_0001, _0002, etc.).

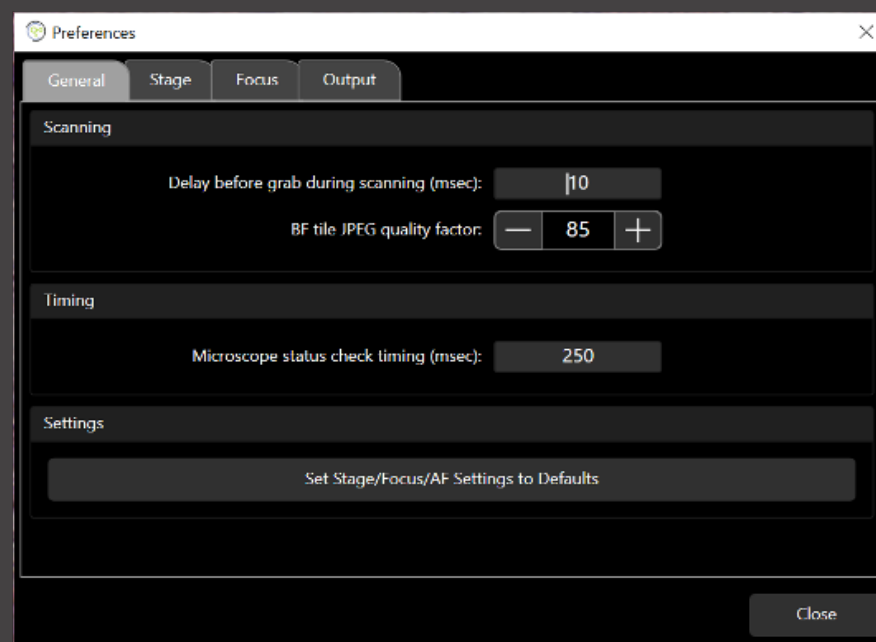
4.3.2.5 System Information

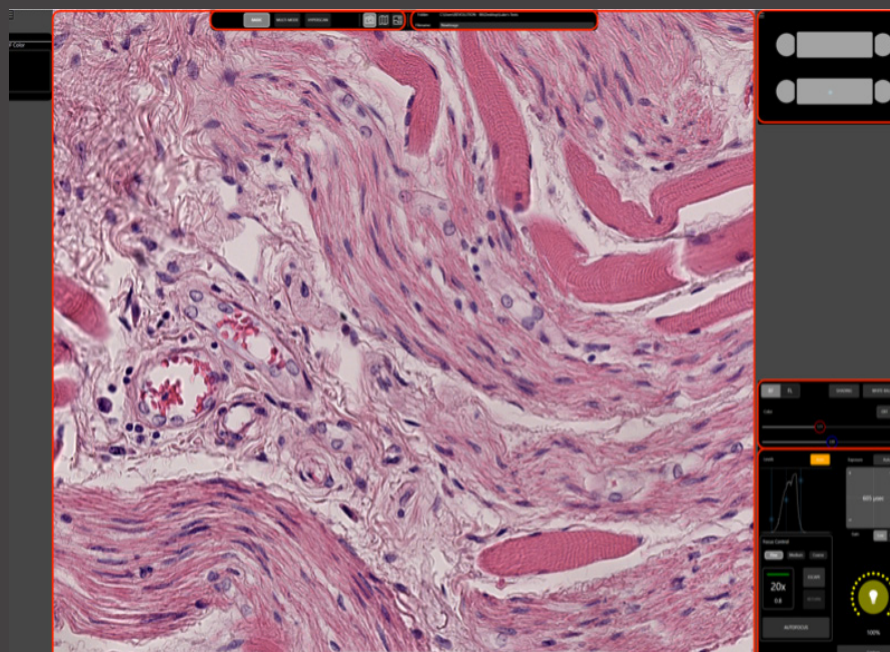
You can get information about your Revolution by clicking the About tab. When submitting a support ticket or request to ECHO, please include a screenshot of this tab for faster diagnosis of any issues you may be experiencing. Please note that the serial number for your microscope is not accurate on this screen. The correct serial number for your microscope is printed on the sticker which is located on the rear-panel of your Revolution.

Img 7: Screen capture of the Revolution software, version 1.0.26, User Preferences menu.

This can be accessed via the hamburger menu > “User Preferences” and offers multiple tabs of adjustable settings, including:

1. Image Capture Delay
2. Focus Multiplier Adjustment
3. Auto Indexing Experiments



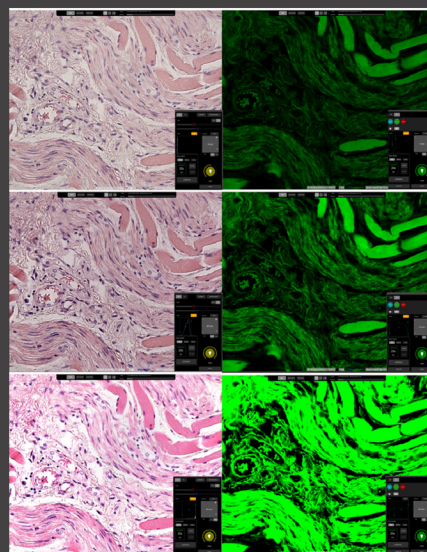


Img 4 Top: Screen capture of the Revolution software, version 1.0.26, in BF Live View.

Various settings menus are highlighted, including the mode selector, stage insert manager, and lighting controls.

Img 8a-f Right: Screen captures of the same section of an HNE slide taken in both BF and FL. The only difference between each image capture were the in the exposure settings.

The top image is under-exposed. Too little exposure may cause "Dark Noise" from the camera, which can appear as small pixelations. The bottom image is over-exposed, which blows out the features of interest.



4.4 BF Basic Mode

Basic mode is the simplest operating mode available on the Revolution. This mode turns off the automated features and allows you to use the Revolution as if it were the Revolve or any unautomated optical microscope.

To start in Brightfield (BF) Basic Mode, select Basic Mode from the Mode Selection Ribbon. Then, select BF in the image settings menu. This will automatically switch you to the Live View (Camera View), which may initially appear blurry, pixelated, or noisy. To solve these issues, please increase your LED brightness and then complete a Shading Correction and White balance. See Section 4.2.1.2 for information on how to complete this.

To navigate around your sample in Basic mode, you can either use the optional Joystick, single-click (or tap) an area of interest to center on that spot, or double-click a location on the sample holder to fly directly to that location on your slide.

To adjust the exposure for a different location on your sample, you can either change the LED brightness, the image exposure time, the levels scaling in your histogram (Levels Table). Changing the LED brightness and image exposure time will change the recorded data values, while changing the levels scaling only effects what is displayed. Your levels table should be set to AUTO as evidenced by the

AUTO button being orange. To turn off auto-levels, click the AUTO button so it appears grey. With the levels in manual mode, you can now adjust the maximum and minimum display value in addition to the gamma (50%) level.

To specify what an image should be called and where it should be saved, utilize the ribbon at the top of the screen. There is a path for the folder, and a textbox for the desired name.

Once a region has been selected for imaging, is in focus, and has been correctly exposed, you are ready to take a picture of your sample. Click the capture button. The picture will be automatically saved to the defined folder path with the specified name. You can either view your image within Revolution by clicking the image from the banner on the left-hand side of the screen, or by navigating to it in Microsoft File Explorer. To return to the image capture window, select Live View.



4.5 FL Basic Mode

Basic mode is the simplest operating mode available on the Revolution. This mode turns off the automated features and allows you to use the Revolution as if it were the Revolve or any unautomated optical microscope.

To start in Fluorescence (FL) Basic Mode, select Basic Mode from the Mode Selection Ribbon. Then, select BF in the image settings menu. This will automatically switch you to the Live View (Camera View), which may initially appear blurry, pixelated, or noisy. To solve these issues, please increase your LED brightness and then complete a Shading Correction. See Section 4.3.1.3 for information on how to complete this.

To navigate around your sample in Basic mode, you can either use the optional Joystick, single-click (or tap) an area of interest to center on that spot, or double-click a location on the sample holder to fly directly to that location on your slide.

To adjust the exposure for a different location on your sample, you can either change the LED brightness, the image exposure time, the levels scaling in your histogram (Levels Table). Changing the LED brightness and image exposure time will change the recorded data values, while changing the levels scaling only effects what is displayed. Your levels table should be set to AUTO as evidenced by the

AUTO button being orange. To turn off auto-levels, click the AUTO button so it appears grey. With the levels in manual mode, you can now adjust the maximum and minimum display value in addition to the gamma (50%) level. You can adjust the exposure per-channel.

To add channels (4 channels maximum + transmitted light), click on the Edit button in the left-side ribbon, or click the ellipses from the ribbon directly below the FL mode selector. You can re-order the channels by dragging them up and down in this menu.

Additionally, you can also toggle pseudo-color and the oversaturation indicator by using the icons located in that same ribbon. Pseudocolor must be enabled to take a colored FL image. You cannot toggle pseudocolor per FL channel. However, focus can be toggled per channel, and in FL Basic Mode, focus is sticky for each channel.

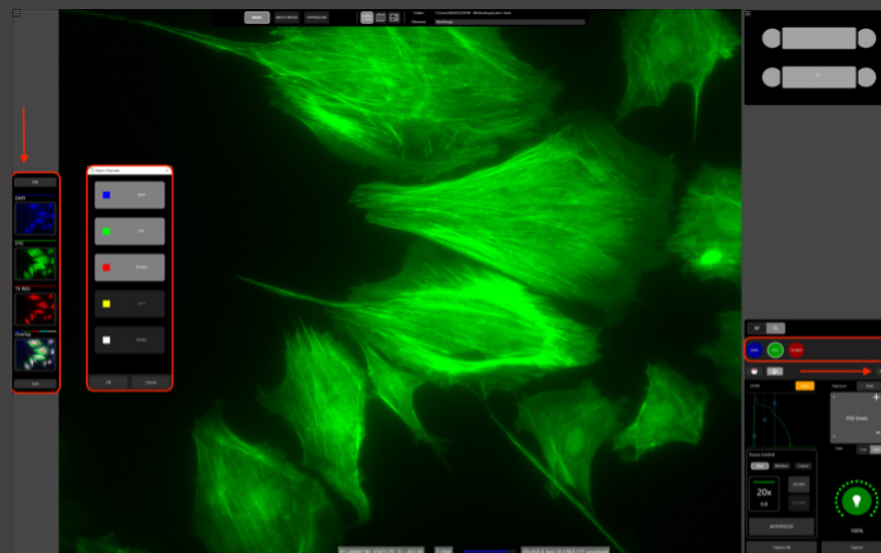
To specify what an image should be called and where it should be saved, utilize the ribbon at the top of the screen. There is a path for the folder, and a textbox for the desired name.

Once a region has been selected for imaging, is in focus, and has been correctly exposed, you are ready to take a picture of your sample. Click the capture button to capture a single channel, or the capture all button

to capture and overlay all channels. The pictures will not automatically be saved to the defined folder path with the specified name. Clicking save will bring up a prompt to select where your image should save to!

Img 9: Screen capture of the Revolution software, version 1.0.26, in FL Live View.

The main differences between FL Live View and BF Live View are highlighted. FL mode has the additional ability of being able to add or remove channels, up to 5, to create a composite image. Channels can be added or removed via the “Edit” or “...” buttons.



4.6 Multimode

Multimode unlocks your Revolution's full suite of automated features:

1. Autofocus
2. Timelapse
3. Multi-point
 - a. Multi-point
 - b. Mosaic
4. Multi-channel – only available in FL mode.
5. Z-stack
6. Focus Map – only available with the purchase of either the HyperScan or Multi-Well Points Module software upgrades.

To create a Multimode experiment, click on the Multimode option in the Mode Selection Ribbon. This will add the Experiment Designer Window to the right-hand menu. To select which automated operations you want on your experiment, click the ellipses button, then click on which features you'd like to add to your experiment. The highlighted features will be included.

The order of operations you choose for the features are important! You can re-order operations by dragging them left or right. Note that the Timelapse operation must be in the primary channel.

Operations are carried out from left to right. For example, if you have Multi-point (MP) and Multi-channel (MC) will take an image with each channel at a single point before moving on to the next point, while flipping these around will have each point imaged by a single

channel before returning to the first point and imaging again with the next channel.

You can run your experiment by clicking Run Experiment. While a Multimode Experiment (Experiment) is running, you can abort at any time by clicking Abort Experiment. When your experiment is complete, click on the Player to view your captured images, along with the option to export them as .jpg, .tiff, or as an .ome.tiff. They will be exported to the file path you specified at the start of the experiment. Images will have your experiment name as a prefix.

4.6.1 Timelapse

When running a timelapse, it is very important to ensure that there are minimal environmental vibrations, otherwise, the images may not correctly overlay. Therefore, ECHO recommends using a vibration table or vibration plate. To add timelapse to your experiment, select Timelapse in the Experiment Designer Window. Underneath the Timelapse tab, you can then set the interval between images and the duration of your timelapse in seconds, minutes, or hours. The software will then determine how many cycles are required to take a timelapse of the desired length.

While there is no maximum length for your timelapse within the software, there is a storage limitation. All data is written to your C:/ drive while the experiment is running, and therefore, the maximum length is determined by

how much room is left on your drive. If you need more space, we recommend moving any other installed programs to an external HDD or SSD.

Note that if the image capture time, including time to focus and move between defined points, is longer than your interval time, the duration will take longer than specified.

4.6.2 Multi-point

There are two options within the Multi-point operation. These are Mosaic and Multi-point. They cannot be used in combination.

4.6.2.1 Mosaic

With Mosaic, your Revolution will take multiple images and stitch these together for you. Other software programs may call this a tiled image.

There are two methods by which you can set your bounds for a mosaic image:

1. Boundaries: Drag a window over the map, and the Revolution software will try to subdivide the selected area with a tiling pattern. However, the boundaries method will not capture 100% of the defined area due to the overlap between bordering tiles and the chip size.
2. Centering: Navigate to the region you'd like to have as the center of your image. Click the X located between the size selectors to center on your location. The left-hand selector changes your width, while the right-side selector

changes your height.

Within Mosaic, you can decide to autofocus every n number of images. To turn this feature on, click the slider next to the word Autofocus and type in the interval you'd like to choose. When this is selected, the first image will be autofocused.

While a Mosaic experiment is running, you will be taken to Map View automatically and can view the images being stitched together in real time. When this experiment is exported, your stitched image will be exported – no after-export processing necessary.

4.6.2.2 Multi-point

With Multi-point, you can select multiple different points on your screen to take an image of. You can either specify points by clicking Add, which will add your current XYZ position, or by right-clicking the desired location in the Map view.

Multi-point is quite useful for multi-well plates (MWP). We recommend saving MWP templates to allow for easy reloading. See Section 7 for more information on how to save and retrieve MWP templates.

If you have the Multi-Well Points Software Module, you will be able to image multiple times at and around each point. You can either create a larger composite image (tiled image), or take multiple images around your desired point.



4.6.3 Multi-channel

To utilize the multi-channel operation, you need to be in FL mode. However, unlike in FL Basic Mode, you need to manually set your z-offset from your priority channel.

Adding channels with this operation works exactly like adding channels in basic mode. You can either add new channels by clicking Edit or by clicking the Ellipses located below the FL selector button.

To set the focus offset for each channel, find the focal plane for your current channel, then hit the Set button underneath the MC operations menu. Move to your next channel and repeat. The exposure and brightness of each channel will also be saved.

4.6.4 Z-Stack

When utilizing both multi-channel and z-stack, it is important to set your Top and Bottom images based off your priority channel. If you are imaging only a single channel or are using Brightfield mode, then ignore this warning.

To set the Top of your z-stack, scroll just above your image's focal plane, so either none of the sample, or just a small portion of the sample, are still in focus. Then click Set. Repeat this for the bottom, but scroll in the opposite direction.

Based off the recommended step size for your objective, the Revolution

software will determine how many images are needed to generate your z-stack. You can change this step size, or the number of images taken for your z-stack.

There are three different options for z-stack: capture each individual focal plane, or generate an MIP or EDF. Each option is mutually exclusive. Selecting the toggle for MIP will disable the toggle for EDF. Having both off will capture and save images of each individual focal plane.

4.6.5 Focus Map

Focus mapping is only available for users who have purchased either the HyperScan or Multi-Well Points Module software upgrades.

To utilize focus mapping, you can either set a point at your current location by clicking "Add", or by adding a point to a location on an already-captured mosaic image. A common experimental procedure utilizing focus mapping is described below:

1. Create a mosaic image with a low magnification.
2. Navigate to Map View and select a sub-region of what was just imaged. This sub region will be automatically named Scan 1 by the program.
3. Right-click on an area of your image with something of interest. This will create a focus point on this location. The GUI will show a red

dot, meaning that the region has not been focused.

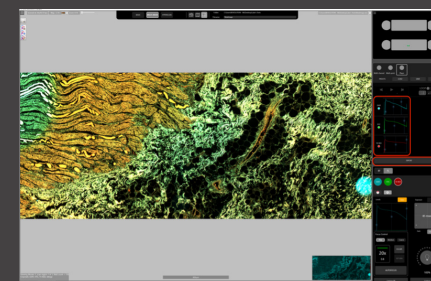
4. Repeat step 3 as many times as needed. It is recommended to have at least 3 points per sample to create an accurate focus map. A maximum of one focus point per tile is recommended.
5. To autofocus each point, select the Focus Map operation tab > AF All. The program will now attempt to autofocus each point, and successfully autofocused points will now appear green in map view.

Focus mapping can be used in conjunction with other operations. However, focus maps do need to be set according to your priority channel. If utilizing a z-stack with a focus map, the focus map should be set to the best focal plane.



Img 12: Screen capture of the Revolution software, version 1.0.26, in FL Multimode Live View.

Underneath the Multi-channel tab, users can set the z-offsets between channels by clicking set. Channels can also be added, removed, or re-ordered by clicking the ellipses.



Img 13: Screen capture of the Revolution software, version 1.0.26, after a Multi-channel mosaic capture.

Post-capture, individual channels can be edited to increase or decrease the displayed contrast between gray levels. Additionally, individual channels can be turned off.



4.7 Hyperscan

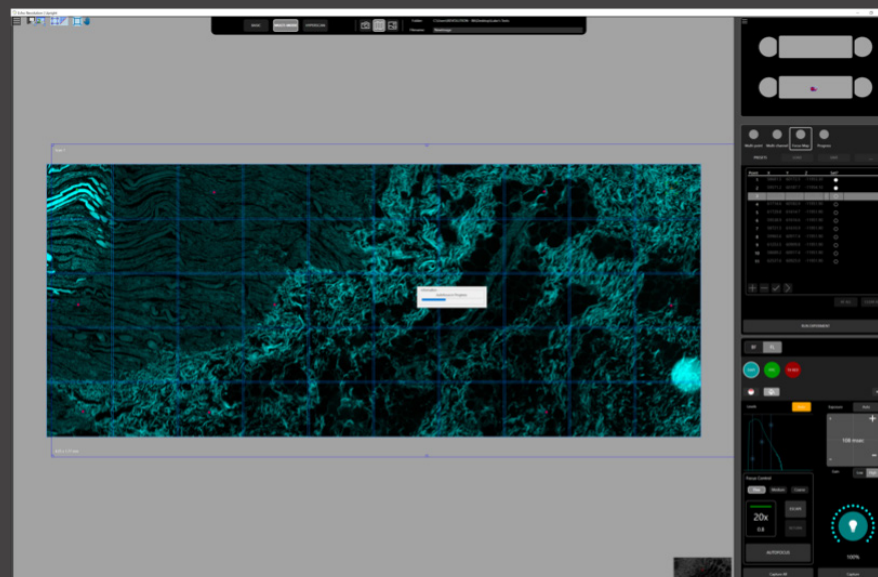
The Hyperscan menu is only available to users who have purchased the HyperScan Software add-on. This menu allows users to adjust a few settings relating to stage speed and image capture, and tiling pattern.

There are two scan modes available in the latest software version, which as of the time of publication (03-20-2023), is version 1.0.26.2. These scan modes are Standard Scan and Hyperscan Stepped. Standard Scan is the default mode for all systems, and is a slower operating mode, while Hyperscan Stepped will take advantage of the full speed setting. Depending on the speed setting and imaging time, it is possible that Hyperscan Stepped may be slightly too fast for your stage, and result in incorrectly stitched images.

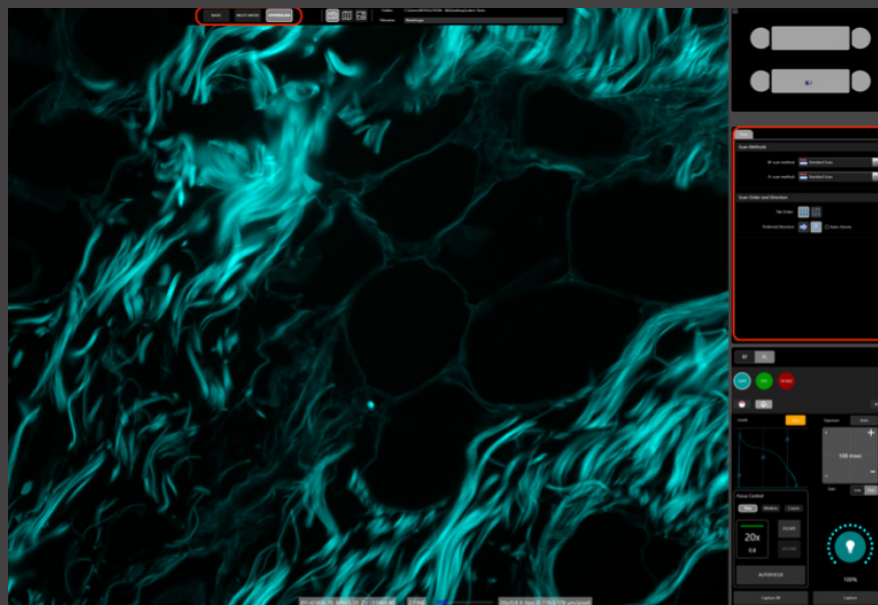
The default tiling order is to start in the top left corner, continue along the top row, then reset to the left-most tile in the next row. To change the tile order to bounce back and forth, click [icon]. You can also change the scan direction to top-down (vertical lines), or have the software automatically decide for you.

Img 14: Screen capture of the Revolution software after completing a Multi-channel mosaic capture.

Focus points have been dropped into various tiles, and the system is now autofocusing each of those locations.



Img 15: Screen capture of the Revolution software, version 1.0.26, in the HyperScan settings Menu.



5. Microscopy Modes

Your Revolution supports 5 different microscopy techniques in both the upright and inverted configurations. Sections 5.1-5.5 will cover how to perform each type of microscopy on the Revolution, while Section 5.6 will demonstrate how to convert your Revolution from Upright to Inverted mode. Each section will also contain a few troubleshooting steps for commonly noted image capture errors.

5.1 Brightfield Microscopy

To take a Brightfield image, please ensure that your condenser is set to “BF” and that your Revolution software is set to use the “BF” camera.

1. Open your condenser aperture fully.
2. In the Revolution software, set the LED intensity to between 30-50%.
3. Locate and focus on your sample. If your sample appears overexposed, please lower your LED intensity and exposure time. If your sample still appears dark, increase your exposure time.
4. If you are in basic mode, click the Capture Button to take your image. If you are in Multimode, confirm that your experiment design is complete before selecting Run Experiment.
5. Save or export your images!

If at any time while taking a Brightfield image you notice that your image lighting is inconsistent, or your image

has a tinge to it, please follow the steps outlined in Section 4.3.1.3 to set a Shading Correction and White Balance. If your Live Image Window appears dark but your sample is being overexposed, enable Auto Levels.

5.2 Fluorescence Microscopy

To take a Fluorescence image, please ensure that your condenser is set to “BF” and that your Revolution software is set to use the “FL” camera.

1. Open your condenser aperture fully.
2. In the Revolution software, select your Fluorescence channel of interest. If taking images of multiple channels for an overlayed image, select your primary channel first. Set the LED intensity to between 50-100%.
3. If you are interested in taking a pseudocolor image, enable pseudocolor now. Please note that pseudocolor cannot be toggled on a per-channel basis.
4. Locate and focus on your sample. If your sample appears overexposed, please lower your LED intensity and exposure time. If your sample still appears dark, increase your exposure time. You can confirm your sample is not being overexposed by utilizing the saturation meter. If you have enabled pseudocolor, the oversaturated regions will appear white. If you have not enabled pseudocolor, the oversaturated

regions will appear red.

5. If you are in Basic Mode and interested in taking an image of several channels, go over to your next channel and repeat the instructions from the previous step (4). Focus, LED intensity, exposure time, and levels are all sticky/channel in basic mode. However, if you are in Multimode, you will need to enable the Multichannel operation and manually set your focus offsets. However, your LED intensity, exposure time, and levels are still sticky/channel.
6. If you are in basic mode, click the Capture Button to take an image of the currently displayed channel, or Capture All to capture all channels and overlay them. If you are in Multimode, confirm that your experiment design is complete before selecting Run Experiment.
7. Save your images. Saving your images in Basic Mode will save all your channels and overlayed image in color. Exporting your images in Basic Mode will save all your channels in grayscale, but your overlayed image in color. Exporting channels from Multimode will export them in grayscale, but the exported overlayed image will be color.

If at any time while taking a Fluorescence image that your image lighting is inconsistent, please follow the steps outlined in Section 4.3.1.3 to set a

Shading Correction. If your Live Image Window appears dark but your sample is being overexposed, enable Auto Levels.

5.3 Phase Contrast Microscopy

Depending on if you are using the ELWD Universal Condenser or High-Resolution Condenser, the instructions on how to perform phase contrast will be slightly different. Please confirm which condenser you have loaded on your microscope before performing phase contrast; and ensure that the condenser has been properly Kohlered. For both condensers, your software will need to be set to “BF” mode.

5.3.1 ELWD Universal Condenser

1. Open your condenser aperture fully.
2. In the Revolution Software, set the LED intensity to 100%.
3. Rotate the ELWD Condenser selector to match your objective. Use of incompatible phase rings will create a distorted image.
4. Load your sample and focus your objectives. Change the LED intensity and exposure time as needed.
5. If you are in basic mode, click the Capture Button to take your image. If you are in Multimode, confirm that your experiment design is complete before selecting Run Experiment.
6. Save or export your images!



5.3.2 High-Resolution Condenser

1. Open your condenser aperture fully.
2. In the Revolution Software, set the LED intensity to 100%.
3. Rotate the High-Resolution Condenser selector to match your objective. Use of incompatible phase rings will create a distorted image.
4. If utilizing an oil immersion objective, apply the proper amount of oil to your objective.
5. Load your sample and focus your objectives. Change the LED intensity and exposure time as needed.
6. If you are in basic mode, click the Capture Button to take your image. If you are in Multimode, confirm that your experiment design is complete before selecting Run Experiment.
7. Save or export your images!

5.4 Darkfield Microscopy

Darkfield microscopy can be performed with on two different condensers offered by ECHO. Before use, ensure that the condenser has been properly Kohlered.

5.4.1 High-Resolution Universal Phase Turret

1. Open your condenser aperture fully.
2. In the Revolution Software, set the LED intensity to 100%.
3. Set the phase selector to DF. The phase rings will block enough light for a lens of below 0.65NA to create a darkfield environment.
4. Load your sample and focus

your objectives. Change the LED intensity and exposure time as needed.

5. If you are in basic mode, click the Capture Button to take your image. If you are in Multimode, confirm that your experiment design is complete before selecting Run Experiment.
6. Save or export your images!

5.4.2 Darkfield Oil Condenser

If you are unsure on how to use an oil condenser or oil lenses, please contact your manager, trainer, or laboratory manager for proper instructions and training.

1. Open your condenser aperture fully.
2. Apply oil to your condenser lens.
3. Load your sample.
4. Raise the condenser so the oil creates a firm seal between the condenser and slide.
5. In the Revolution Software, set the LED intensity to 100%.
6. Focus your objectives. Change the LED intensity and exposure time as needed.
7. If you are in basic mode, click the Capture Button to take your image. If you are in Multimode, confirm that your experiment design is complete before selecting Run Experiment. If tiling, it is recommended to use a clip to prevent your slide from sticking to the condenser due to the viscosity of the oil.
8. Save or export your images!

5.5 Polarized Light Microscopy

ECHO offers two different polarizer options – a linear polarizer and a circular polarizer. Both polarizers are operated in the same way.

1. Place the polarizer over the condenser.
2. Offset the two by 90°.
3. Load your sample and focus your objectives. Change the LED intensity and exposure time as needed.
4. If you are in basic mode, click the Capture Button to take your image. If you are in Multimode, confirm that your experiment design is complete before selecting Run Experiment.
5. Save or export your images!

5.6 Revolving your Revolution

Failure to follow these steps could result in damage to your microscope or objectives.

To revolve your microscope:

1. Retract your objectives as far away from the stage as possible.
2. Remove your sample.
3. Slide the stage:
 - a. If rotating from Upright to Inverted, slide your stage to the Left.
 - b. If rotating from Inverted to Upright, slide your stage to the Right.
4. Push the long, skinny button located on the top of your microscope body. The microscope will start to rotate on its own.
5. Rotate the body to the desired configuration (upright or inverted). Once in place, you will feel, and possibly hear, a firm click. This means the body is now locked in place.
6. Replace the stage to the central position.
7. Replace your sample.
8. Focus your objectives.

The Revolution does not have a halfway mode between upright and inverted (perpendicular to benchtop).



6. Technical Specifications

Failure to adhere to the power and environmental requirements can result in damage to your equipment, or personal injury. ECHO is not responsible for damages caused by improper usage of equipment.

6.1 Power Requirements

Your system will come with a power adapter suited for your country's or municipality's standard outlet and power supply type. Failure to utilize the ECHO-supplied power connectors may result in equipment damage.

The Revolution has following power requirements:

Input Voltage: 100-240V 50/60Hz
 Input Current: 2.0A
 Output Voltage: 24V
 Output Current: 6.67A
 Output Power: 160W (Max)

The Surface Studio has the following power requirements:

Input Voltage: 100-250V 50/60Hz

The Revolution Incubator has the following power requirements:

Input Voltage: 12V
 Input Current: 9.4A (Max)
 Voltage Fluctuation: +/- 10%
 Overvoltage Category 2, Pollution Degree II, Altitude up to 2000m.

The FC-5N Automatic CO2/Air Mixer has the following power requirements:

Input Voltage: 100-200V 50/60Hz

6.2 Environmental Requirements

The Revolution, Computer, and Revolution Incubator have the following Environmental Requirements:

Operating Temperature: 0° to 104°F (0° to 40°C)

Storage Temperature: -4° to 113°F (-20° to 45°C).

Relative Humidity: 5% to 95% noncondensing.

6.3 External Storage

The Revolution supports external storage drives via USB connection using the included Windows 10 Computer. However, to ensure that all file sizes can be saved to your external drive, we recommend using NTFS or exFAT. In particular, exFAT formatting is recommended for users whose facilities utilize computers from a wide-range of operating systems, such as Mac or Linux.

If the computer does not recognize your thumb drive or external drive, please ensure that you are using a high-quality drive and that it is correctly formatted. If your external drive (SSD, HDD) is not detected, please utilize a drive with an external power supply.



7. System Troubleshooting

7.1 Unresponsive Revolution Software

If your Revolution Software freezes or crashes during an experiment, please contact ECHO Customer Support with your Serial Number, Software Version, and what you were doing at the time of the crash. Note that rebooting the Revolution Software will erase any data that has not yet been saved to the hard drive, and will also end any timelapse experiments you are currently running.

To reboot the software, open the Task Manager on your computer (Ctrl-Alt-Delete > Task Manager), and click the Revolution Line. Select End Program. If you cannot open Task Manager as your computer has also frozen, hold down the power button to perform a hard reboot of your entire computer system.

7.2 Software Freezes on Bootup

If your Revolution Software freezes or crashes during an experiment, please contact ECHO Customer Support with your Serial Number and Software Version.

7.3 Rebooting the Microscope

To reboot your Revolution, close the Revolution software, then unplug the microscope's power cord from the wall. Wait 30 seconds, then plug it back in. Wait another 30 seconds before opening the Revolution software.

7.4 Camera Connection Failure

If your screen appears black, you get a notification stating that the USB connection has been disconnected, or a notification that the BF and FL cameras have been disconnected:

1. Unplug your Revolution power cord from the wall.
2. Disconnect the USB connector from your computer and the microscope. Flip the cable the other way.
3. Plug your Revolution back in.

If this does not solve your issue, contact ECHO Customer Support.

7.5 Cannot See Image

If the LED is on but you cannot see your image, there are a few things you can try. If these tips don't work, please follow the steps outlined in Sections 7.1 and 7.3.

1. Make sure your sample is in focus.
2. Increase the LED intensity
3. Increase the exposure time. Tap the center of the image to auto-expose the image.
4. Increase the aperture opening size.
5. Verify that the condenser ring has the proper phase ring in conjunction with the selected objective.

7.6 Image Quality is Poor

If your image quality is poor (ex. dark, pixelated, grainy), one of the following steps may help improve the quality.

1. Check to see if the coverslip is facing the objective.
2. Check to see if your selected lens is LWD (Long Working Distance) or coverslip corrected.
3. Check your microscope condenser settings. Verify that if you are taking a Brightfield or Fluorescent image, that "BF" is selected on the condenser.
4. Check that your aperture is open.
5. Check that your samples are loaded correctly and seated firmly.
6. Adjust the exposure settings (brightness, white balance, etc.) within the Revolution Software.

7.7 Sample Not Appearing Flat

If your sample does not appear flat, it is possible that your sample may not be smooth. However, if your sample has a noticeable tilt, you can reduce the effect by:

1. Checking that the stage is correctly latched.
2. If you are using a slide, check to see that the slide coverslip is in the upright position, not inverted.

7.8 Fluorescence Channel Disappeared

If one or more of your fluorescent channels have disappeared from the FL window, it's likely that they were disabled for another user's experiment. To re-enable your FL channels:

1. Select the "..." in the FL menu.
2. Select the channels you wish to activate.

It may also be possible that the channel was renamed. To rename and recolor, right-click the channel, then enter in a new name. Please note that names cannot contain Windows-reserved characters.



7.9 Fluorescence Channel Not Activating

If one or more of your fluorescence channels are not activating, try the following:

1. Adjust the exposure time of that channel. Some channels may require more LED light than others to show up.
2. Ensure the LED and filter cube are correctly matched. For example, if your DAPI LED is in position 1, the matching DAPI filter needs to be in position 1.

It may also be possible that the channel was renamed. To rename and recolor, right-click the channel, then enter in a new name. Please note that names cannot contain Windows-reserved characters.

7.10 Fluorescence Channel Not Activating

If one or more of your fluorescence channels are not activating, try the following:

1. Adjust the exposure time of that channel. Some channels may require more LED light than others to show up.
2. Ensure the LED and filter cube are correctly matched. For example, if your DAPI LED is in position 1, the matching DAPI filter needs to be in position 1.

If the previous steps did not resolve your issue, perform a light box test.

1. Revolve your microscope into the inverted configuration.
2. Switch to an open position on your nosepiece (no objective). Place a flat piece of white paper, a kimwipe, or a tissue over the hole.
3. Open the Revolution Software and toggle your first fluorescent channel. Raise the LED brightness to 100%.
4. Confirm that a square appears on the sheet and that it has bright, crisp edges.
5. Switch to the next channel and repeat.

If one of your channels is not square, bright, or crisp, you will need to have your light box replaced. Contact ECHO Customer Support or your ECHO-Approved Vendor for help.

7.11 Fluorescence Channel Out of Focus

If a single, or multiple, fluorescent channel appear out of focus in your completed Multimode experiment, despite appearing in-focus in the live view window, it is likely that you did not set your offsets correctly.

To set your z-offsets:

1. Find the focal plane for your primary channel.
2. Navigate to the MC operations menu, then hit “Set” next to that channel. This sets the focal plane for your current channel to your current z-plane.
3. Move to your next channel and repeat.

7.12 Sample is Unevenly Illuminated

Uneven illumination can occur in BF or FL if your camera has not had a shading correction. To conduct a shading correction, please check section 4.3.1.

If your sample is unevenly illuminated when conducting PC, please repeat Kohler illumination.

7.13 Repeated Shading while Tiling

When taking a tiled image, shading can become noticeable as the same pattern repeats in each of the tiled sections. The Revolution has three different types of shading issues resulting from tiling: Bleaching, Shading, and Patching.

7.13.1 Shading

Repeated shading can be resolved via a shading correction and white balance. Please check section 4.3.1. for information on how to do this.

7.13.2 Repeated White Lines

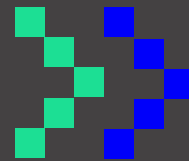
Repeated white lines occur when the stage is moving too quickly for image overlap to occur. The best ways to resolve this are increasing the time delay for when an image is captured underneath user settings, or increasing your exposure time and decreasing the LED intensity

7.14 Focal Drift

If your z-plane shifts slightly after focusing, it is possible that possible that your motor may be quite sensitive. This can be remedied with a software adjustment:

1. Open the hamburger menu in the top left.
2. Select “Preferences...”
3. Select the “Focus” tab.
4. Decrease the Focus Closed Loop Factor (Encoder Counts) value. The lowest value this field can have is 1 (one).





For more information about your Rebel, or any help with Setup and Operation, please contact us on our website at discover-echo.com.

